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B.Tech. Degree VIII Semester Regular Examination April 2023

ME 19-205-0802 MATERIALS MANAGEMENT

(2019 Scheme)

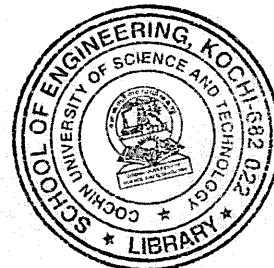
Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the scope, objectives and phases in material management.
 CO2: Get concept of static inventory problems.
 CO3: Study dynamic inventory problems under risk.
 CO4: Learn about lot sizing in material requirement planning.



Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

PART A (Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	What are the three phases in Materials management? Explain any two methods of storage.	3	L1,L2	1,2	1
(b)	Explain with example (i) Invoice (ii) Tender.	3	L1,L2	1	1
(c)	Explain Opportunity Cost and Cost of Risk.	3	L1,L2	2	1
(d)	Derive EOQ formula, stating assumptions.	3	L1,L2,L3	2	1
(e)	Explain P system of Inventory Control.	3	L1,L2	3	1
(f)	What are the defining characteristics of dynamic inventory problems under risk? Name any one probability distribution that is used to find out optimum level of inventory in dynamic inventory problems under risk.	3	L1,L2	3	1
(g)	What is Product Structure? Explain with an example.	3	L1,L2,L3	4	2
(h)	What are the inputs required for Material Requirements Planning? What is Inventory Stock File?	2 1	L1,L2	4	1

PART B

(4 × 12 = 48)

II. (a)	What is Vendor Rating?	1	L1,L2,L3	1	1
	What are the parameters considered for Vendor Rating?	2			
	Explain the Procedure of Vendor Rating.	3			
(b)	What are the objectives of keeping Inventory?	2	L1,L2,L3	1	1
	What are the different types of Inventory Problems?	2			
	What are the relevant costs of keeping the Inventory?	2			
OR					
III. (a)	What are the different costs considered for analysing Inventory Problems?	3	L1,L2,L3	1	2
	With an example, explain the methodology of analysing the Inventory Problems to find out optimal level of inventory.	3			
(b)	Explain ABC, VED and FNSD analysis of classification of inventory. Under what conditions each is useful?	6	L1,L2,L3	1	2

(P.T.O.)

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- | | Marks | BL | CO | PO |
|---|-------|----------|----|----|
| IV. (a) Mathematically formulate the inequality to solve continuous case of Static Inventory Problems under risk. | 5 | L1,L2,L3 | 2 | 3 |
| (b) Consider the inventory model for 2 items with the following data: | 7 | L3,L4,L5 | 2 | 4 |

	Item 1	Item 2
Annual demand	11,000	6,000
Order cost (₹/order)	350	325
Inventory carrying cost (₹/unit/year)	2	2

There is a constraint that the number of orders should not exceed 10 per year for the items together. Find the optimal order quantities and the corresponding cost.

OR

- | | | | | |
|--|---|----------|---|---|
| V. (a) A company is making a large boiler installation. A certain automatic monitoring unit is crucial for the operation of the whole system. The original spares for this unit can be purchased for ₹2,000 each. The probability distribution for failures of this unit during the life time of the installation is known to be | 5 | L3,L4,L5 | 2 | 3 |
|--|---|----------|---|---|

Demand:	0	1	2	3	4
Probability:	0.30	0.25	0.25	0.10	0.10

If a spare is needed and is not available, the total cost to the company in down time and replacement cost will be ₹15,000. Unused spares have no salvage value. Formulate Total cost matrix and determine the optimum number of spares to stock.

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|---|---|-------|---|---|
| (b) Find the optimal order size for a product where the annual demand for the product is 500 units, the cost of storage per unit per month is 2% of the unit cost and ordering cost is ₹180.00. The unit costs have the following price breaks; | 7 | L3,L4 | 2 | 3 |
|---|---|-------|---|---|

$0 < Q1 < 500$	₹25
$499 < Q2 < 1500$	₹24.80
$1499 < Q3 < 3000$	₹24.60
$2999 < Q4$	₹24.40

- | | | | | |
|--|---|----------|---|---|
| VI. (a) Explain the characteristics of Dynamic Inventory Problems Under Risk. | 3 | L1,L2 | 3 | 1 |
| (b) For an item the annual average demand is 1000 units. Standard deviation of demand per week is = 10 units. Unit cost ₹6/- Ordering cost ₹11 per order. Inventory carrying cost 12%. Service level 98%. Find Safety Stock and Maximum Inventory level and total cost for "P" system of inventory control using exact method. | 9 | L3,L4,L5 | 3 | 2 |

OR

(Continued)

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Marks BL CO PO

- VII. (a) Explain Two Bin system of Inventory Control. 3 L1,L2 3 3
 (b) An item has a six month selling period with the following probability distribution of selling price in each month. 9 L3,L4,L5 3 4

Price	M1	M2	M3	M 4	M 5	M 6
20	0.20	0.15	0.10	0.05	0.05	0.05
21	0.15	0.15	0.10	0.10	0.25	0.25
22	0.10	0.10	0.30	0.15	0.30	0.25
23	0.05	0.10	0.20	0.25	0.20	0.25
24	0.10	0.15	0.15	0.20	0.10	0.10
25	0.15	0.20	0.10	0.15	0.05	0.05
26	0.25	0.15	0.05	0.10	0.05	0.05

Formulate the optimum selling policy.

- VIII. (a) An item K is manufactured using two components L and M .To manufacture one unit of K, one unit of L and Two units of M are required. The lead time for the production of K is one week, that of L is one week and that of M is two weeks. The requirement of K in the 4th week is 600 units and in the seventh week is 750 units. On hand inventory of K item is 200 units. Generate MRP report for items K, L and M 7 L3,L4 ,L5 4 4
 (b) Explain any one Lot Sizing method used in MRP with an example. 5 L1,L2 4 4
- OR**
- IX. An item A is assembled with 4 units of B and 2 units of C. B is produced by assembling 2 units of D and one unit of E. C is assembled with 3 units of D and 2 unit of F. Generate MRP report for items A,B,C,D,E, and F. Data given are 12 L3,L4,L5 3 4

Lead Times

A	1 day
B	2 days
C	1 day
D	3 days
E	4 days
F	1 day

Demand

Day 10	50 A
Day 8	20 B (Spares)
Day 6	15 D (Spares)

Lot size

A	L4L
B	4
C	L4L
D	6
E	4

Bloom's Taxonomy Levels

L1 = 20%, L2 = 20%, L3 = 30%, L4 = 25%, L5 = 5%.
